Peter A. Dunson

+1 919-667-7880 | peteradunsonn@gmail.com | Linkedin | GitHub | Website

Experience

Teaching Assistant *Gambier, Ohio, USA*
Kenyon College, Department of Mathematics and Statistics *August 2026 – current*

- Teaching assistant and tutor for MATH 336, Probability. Hold two help sessions per week.

Student Member, Statistics Faculty Search Committee *Gambier, Ohio, USA*
Kenyon College, Department of Mathematics and Statistics *October 2026 – November 2026*

- Served as the student member of the search committee for a new statistics faculty member.

Undergraduate Researcher *Baltimore, Maryland, USA*
Johns Hopkins University, Department of Biostatistics *August 2024 – August 2026*

- Worked with Dr. Ciprian Crainiceanu and Dr. John Muschelli on functional data analysis of wearable accelerometry data from NHANES and the UK Biobank.
- First author on an in-progress comparison of objectively measured physical activity in the US and the UK, built on end-to-end harmonized processing of raw high-resolution wrist accelerometry (5,074 US and 28,095 UK participants under a matched wear rule).
- Developing Bayesian factor model methodology and software, including C++ Gibbs samplers for six model classes and a tutorial paper on Bayesian factor models with Dr. Crainiceanu.
- Areas of research: Functional data analysis, physical activity epidemiology, Bayesian factor models, high-dimensional covariance estimation, statistical computing

Undergraduate Researcher *Ann Arbor, Michigan, USA*
University of Michigan, Big Data Summer Institute *June 2024 – July 2024*

- Six week research intensive program in the Department of Biostatistics, working under Dr. Rahul Ladhania and Dr. Snigdha Panigrahi.
- Predicted in-hospital mortality among ICU-admitted heart failure patients in MIMIC-III using bootstrapped elastic net and random forest for variable selection, and Bayesian profile regression to identify latent patient risk profiles.

Golf Course Maintenance *Durham, North Carolina, USA*
Washington Duke Inn & Golf Club *June 2021 – August 2021*

Education

Kenyon College *Gambier, Ohio, USA*
Bachelor of Arts in Mathematics and Statistics, Concentration in Computing *August 2023 – May 2027*

- Relevant coursework: Real Analysis I (MATH 341), Real Analysis II (MATH 441), Mathematical Statistics (MATH 436), Bayesian Statistics (STAT 306), Probability (MATH 336), Nonparametric Statistics (STAT 216), Statistical Computing in R (STAT 226), Data Analysis (STAT 206).

Available preprints

Dunson, P. (2026), **Bayesian precision factor models for high-dimensional macroeconomic time series**, arXiv:2611.12368, to be submitted to *JASA ACS*.

Dunson, P., and Crainiceanu, C. (2026), **Tutorial for Bayesian Factor Models**, arXiv:2607.11819, accepted and under minor revision at *Journal of Reproducible Statistics*.

Manuscripts in Preparation

Dunson, P., Koffman, L., Crainiceanu, C., and Muschelli, J. (2026). **Comparing objectively measured physical activity in the US and the UK using end-to-end harmonized high-resolution wrist accelerometry processing**. Expected submission: fall 2026.

Poster presentations

Kenyon College December 2026, Gambier, Ohio, USA – Bayesian Precision Factor Models

Kenyon College December 2025, Gambier, Ohio, USA – Disagreement Among AI Models as a Metric of Economic Uncertainty: Testing AI Disagreement Against Traditional Uncertainty Metrics

University of Michigan, July 2024, Ann Arbor, Michigan, USA – Predictive Modeling of In-Hospital Mortality in ICU-Admitted Heart Failure Patients

Software

Dunson, P. (2026) **vapor**: Bayesian Precision Factor Analysis for Vector Autoregressions. GitHub

Dunson, P. (2026) **factorverse**: One stop shop for Bayesian Factor Models. GitHub

Dunson, P. (2025) **renderarxiv**: Terminal tool to search arXiv papers and render them as HTML for humans and LLMs. GitHub

Dunson, P. (2025) **renderscholar**: Google Scholar scraper that pulls papers, ranks them by relevance/citations/recency, and outputs an HTML file with a Human view for browsing and an LLM view for copy-pasting into models. GitHub

Skills

Languages: English (Native), Latin (intermediate reading proficiency)

Honors/Awards

- Kenyon Merit List, 2023-present
- First Team All American in Swimming, 2025, 2026
- First Team Scholar All American in Swimming, 2025, 2026
- CSC Academic All-District, 2026

Organizations

- Varsity Swim Team
- American Statistical Association
- Pi Mu Epsilon (Mathematics Honor Society)

Hobbies and Interests

- Photography
- Piano
- Hiking
- Reading
- Adventurous Swimming